UNIVERSITÉ DU LUXEMBOURG

LUCET | LUXEMBOURG CENTRE FOR EDUCATIONAL TESTING

OASYS 3.6

OASYS Manual for Administrators

30 October 2026

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Chapter 1

Introduction

1.1 About OASYS

OASYS is a multiplatform, multi-device digital testing and assessment system built with **modern web technologies**. It has been carefully **designed around business processes following a research-based HCI approach**, ensuring an optimized user experience.

To ensure secure data collection and a high level of service availability, OASYS provides powerful live monitoring for test administrators. Responses are transferred to OASYS servers during the test. Transfer timing depends on the configuration and network connection; data not yet received by the server may be lost if a client crashes.

OASYS developers receive regular security training, and the system undergoes **penetration testing and code review by external security consultants**.

OASYS records timestamps for logged test actions, making the platform well suited to behavioural analysis and data mining.

Large deployments can use load balancing. Supported concurrency depends on the server architecture, assessment content and workload; validate capacity with representative load tests.

Although OASYS was designed primarily as an online assessment system, the client can also run as an app with a server installation on a laptop and a mobile Wi-Fi access point. This enables digital testing in areas with limited or no internet connectivity.

Since 2010, OASYS has been **used successfully in the Luxembourg school monitoring programme ÉpStan**. Since 2014, it has also been the platform of choice for student course evaluations and a range of in-house surveys at the University of Luxembourg.

1.2 Purpose of this Manual

This manual provides administrators with the necessary information to install, configure, manage, and maintain an OASYS installation. It covers system setup, system maintenance, user management, content creation, troubleshooting procedures to keep the application up to date and backup procedures.

1.3 Target Audience

This manual is intended for individuals who want to:

- Install OASYS on a web server.
- Perform administrative tasks on an existing OASYS installation.
- Maintain or extend an existing OASYS installation.

Parts of this manual require knowledge of web server systems and access to the system on which OASYS is hosted.

Chapter 2

OASYS Installation

First, download the OASYS installer from https://oasys.lu/index.php/download

2.1 System Requirements

OASYS 3.6 requires the following software and runtime settings. Install current security patches within the supported ranges; a minimum version is not a recommendation to keep an old patch release.

- **Apache HTTP Server 2.4**, with current security updates. Confirm the exact Apache patch minimum with the installation-package provider; the runtime check does not establish it.
- **PHP 8.3.23 or later, below 8.6.0** (PHP 8.3, 8.4 and 8.5 branches).
- **MariaDB 11.8.2 or later, below 12.4.0**. Use a maintained release branch.
- Required PHP extensions: **XMLReader, XMLWriter, XML, BCMath, Zlib, ZIP, mbstring, PDO, pdo_mysql, pdo_sqlite, cURL, Fileinfo and OpenSSL**.
- The PHP **LDAP** extension is required when LDAP authentication is used.
- PHP post_max_size must exceed upload_max_filesize.
- PHP max_execution_time must be at least **300 seconds**.
- PHP memory_limit must be at least **256M**; **512M or more** is recommended.

The application database account requires SELECT, INSERT, UPDATE, DELETE, CREATE, DROP, ALTER, INDEX, LOCK TABLES, EXECUTE, CREATE VIEW and SHOW VIEW on the OASYS database. A grant warning can mean the effective privileges cannot be fully enumerated from the application connection; ask a database administrator to verify the host-specific grants.

The database must use utf8mb4 with collation utf8mb4_unicode_520_ci. The system check requires table_definition_cache of at least 1024 and a PHP/database clock difference no greater than 15 seconds.

The OASYS 3.6 system check also requires innodb_flush_log_at_trx_commit=0. This setting trades transaction durability for throughput: recently committed data may be lost after a crash. Agree this tradeoff with your database administrator before changing it.

Provide HTTPS, sufficient storage for media and backups, and a supported browser with JavaScript and cookies enabled. Size CPU, RAM, disk and network capacity for the expected assessment workload; no universal hardware capacity is established here. Test the intended client devices and media before deployment.

Apply the file permissions reported by the installer and system check. The web application needs write access to its operational directories, including customContent, media, logs, backupRestore and, for conventional installations, pkg_installer and oasys_ver.txt. Container deployments have different update and write-access rules.

Upstream maintenance information: https://www.php.net/supported-versions.php and https://mariadb.org/about/. PHP 8.2 is no longer in the verified OASYS range; its upstream security support ends on 31 December 2026. Make sure all necessary PHP extensions are enabled on your server for full functionality. The **OASYS installer will also check for the required settings** during the installation process and notify you of any missing components or misconfigurations.

Important Note

To complete the installation, you must have access to your **database administration account with the required privileges**. This is required so that the installer can:

- Create a **new database**
- Set up a **dedicated database user** with the necessary permissions

Make sure you have these credentials ready before starting the installation.

2.1.1 Encryption

For security reasons, OASYS encrypts sensitive data before storing it in the database.

While backend user logins will have hashed passwords, for highest security, other sensitive data such as the password to an LDAP server are encrypted using the OpenSSL library.

Also the passwords for test takers are encrypted before being stored in the database. Since these need to be editable in the editor and exportable, they cannot be hashed.

After installation, no encryption key is defined, and OASYS will generate a random key the first time it tries to read the configuration.

For this reason it is imperative that **the directory <root folder>/conf/ is writable by the web server user**.

Be aware that you should **back up** the **file <root folder>/conf/crypt.ini** after it has been generated. Without it, parts of your database will be useless and OASYS will not be able to function properly.

Alternatively, you can also define your encryption key in environment variables, by defining the following variables:

- **oasys_aes_key:** The key (32 bytes / 256 bits; 64 hexadecimal characters) written in hexadecimal
- **oasys_aes_iv:** The IV (16 bytes / 128 bits; 32 hexadecimal characters) written in hexadecimal

As with the database credentials, you can also define custom names for these variables by **editing the file <root folder>/conf/env.ini** and changing the values as you see fit.

2.2 Baseline Installer

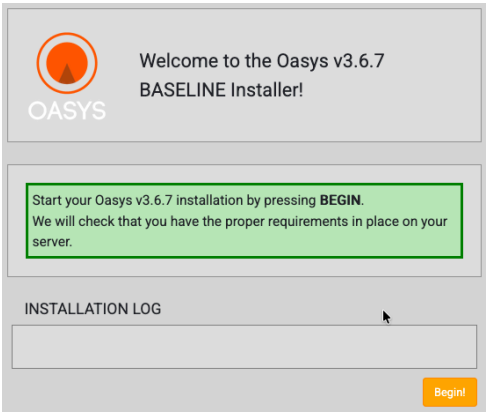
The OASYS installation package comes as a ZIP archive that contains everything needed to get the system up and running on your server.

The Baseline Installer is the installation package for OASYS. After downloading, the installer is uploaded to the web server. The setup is started by accessing the web server directory.
The OASYS installer will guide you step by step through the setup process, including configuration and database setup.
Keeping the system up to date does not require installing a new version. **A system upgrader is integrated into the backend for performing updates.**

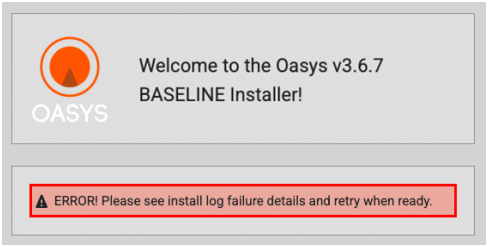
2.3 Installation Procedure

- **Download the ZIP Archive**
  Obtain the OASYS installation ZIP file from the official source or your software provider.
- On your web server, **create a folder** for the OASYS installation.
- **Unpack** the installation archive and **upload** the content to your server folder.
- **Launch the Installer**
  Open your web browser and navigate to the URL where you placed the files. For example:
  https://yourserver.example/oasys/

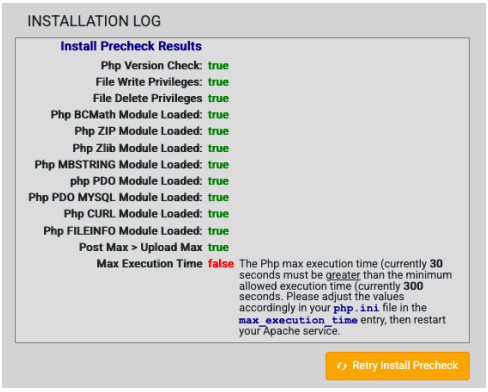
The OASYS installer will guide you through the rest of the setup process, including configuration and database setup.



Click the begin button to start the installation precheck. This will only check the OASYS requirements. No changes will be made.

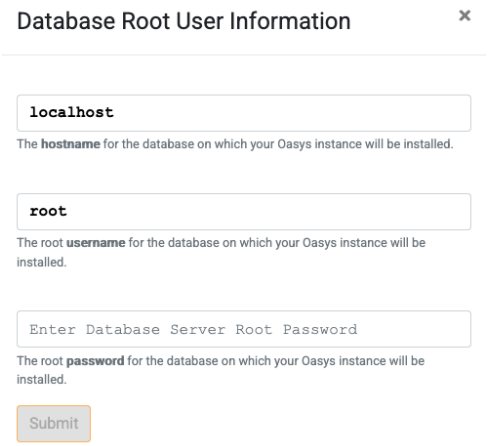


If your system does not meet all requirements, you'll see a warning message.



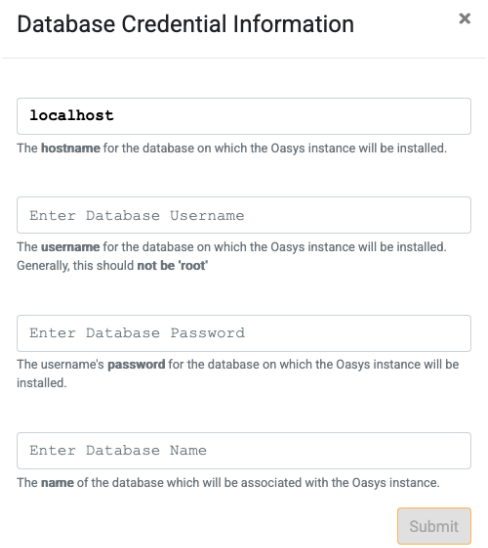
The exact problems are indicated in the log, along with an explanation. Fix them and retry the check.
You will have to restart your web server to apply the changes made!

Once the precheck has passed successfully, continue to the **Root Database Credentials**.

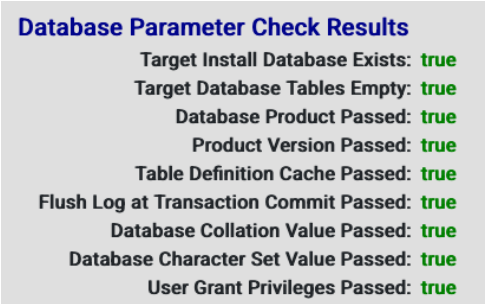


Host and username for the database server are editable.
Be aware, that the user you provide must have the privileges to create a new database and set up a dedicated database user for the OASYS database!

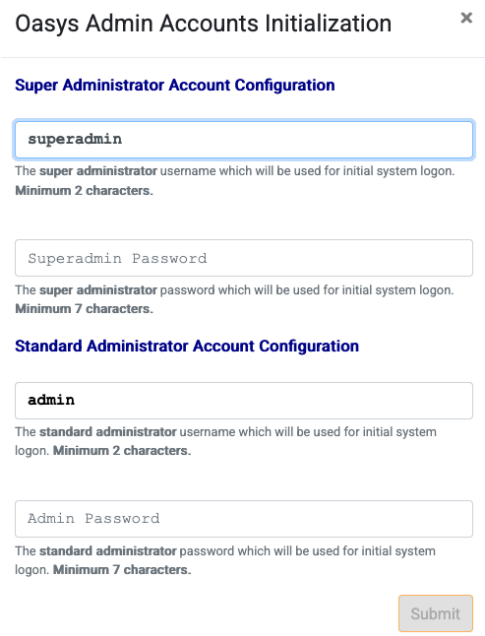
If the user you have provided has the required privileges, click the **User Database Credentials** button.



Enter a username, a password and a database name to be created for your OASYS installation.
If necessary, edit the host.



For the next step, click **Check Database Parameters**.
This will perform a check of the database settings to verify that your database settings meet the requirements for OASYS.
If the settings meet the requirements, click **Start Database Install**.



The next step is setting up initial admin and superadmin accounts.

Super Administrator

- This role is intended to administer the technical part of the OASYS installation.
- Super admins have access to the system variables.
- They can use the upgrader app to update the OASYS version.
- They have the privilege to create admin and superadmin users.

Administrator

- This role is intended to administer users and user groups. This includes setting access permissions for folders.

Submitting this data will create the OASYS database and its basic content. The next step is **installing the file system**.
As the last step, the installer will **validate the installation.**
You can proceed to the backend login by clicking **Go To Editor**

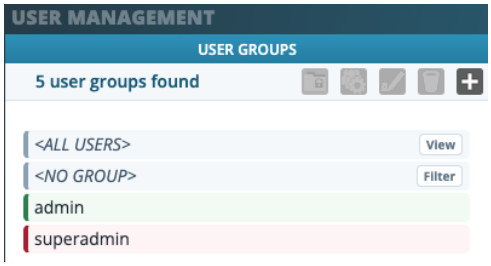
Backend and Frontend

The OASYS frontend URL (login for test takers) is always the URL (and path, if you used a subfolder) you extracted the installer to.
The backend URL is the URL you installed from, followed by **/editor**.

Chapter 3

Overview of OASYS Administration

3.1 Admin Roles and Permissions



There are two predefined user groups in OASYS:

- admin
- superadmin

Users who belong to one of these user groups are granted specific, predefined access permissions.

Note: ALL USERS and NO GROUP are not groups but filters!

3.1.1 Superadmin

The superadmin role is primarily intended for maintaining and configuring the OASYS installation. A superadmin has unrestricted access to all functions of OASYS. The privileges of a superadmin include:

- Creating users in the superadmin and admin group.
- Assigning the role of elevated admin to users in the admin group.
- Bulk editing of folder permissions.
- Permission to delete entries from the log analyzer (maintenance).
- Localization (add new test languages to the OASYS installation).
- Manage and change the system settings (OASYS configuration variables).
- Create backups and restore backups.
- Access to the OASYS upgrader in order to keep the installation up to date.

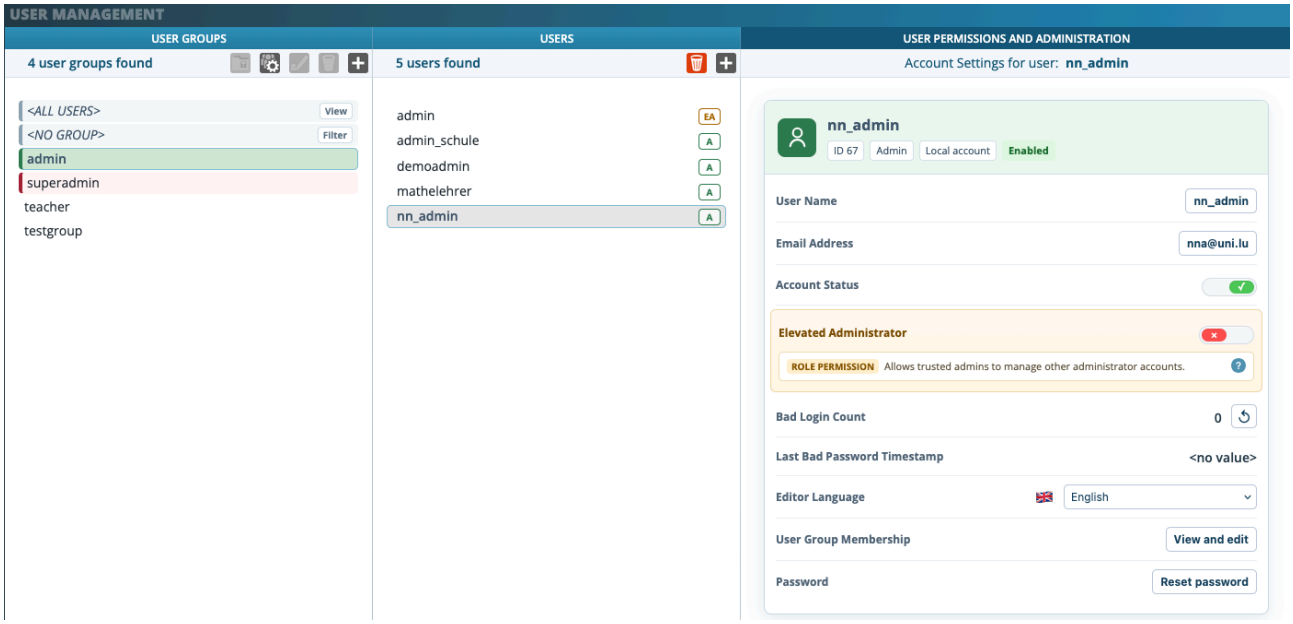
3.1.2 Admin

The default admin role manages ordinary users and user groups. Managing admin accounts requires elevated administrator privileges; managing superadmin accounts requires superadmin privileges. The privileges of an admin include:

- Create and delete user groups.
- Assign custom-module access to user groups.
- Add and delete users.
- Activate and deactivate users.
- Check folder permissions for user groups.

3.1.3 Admin with Elevated Rights

The elevated admin is an admin with extended privileges.



User Details in Superadmin View

For making an admin an elevated admin, you have to be logged in with superadmin privileges!

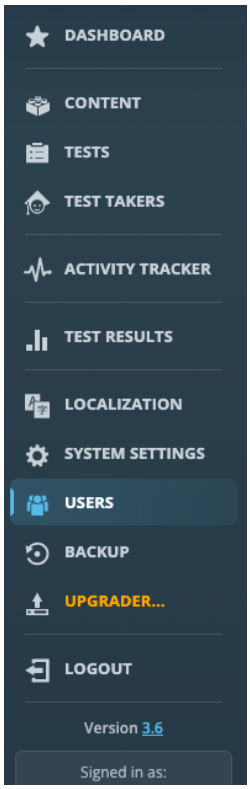
- Select a user from the admin group.
- In the right column, look for **Elevated Administrator**.
- Toggle the **value** to enable the elevated administrator privileges.

The elevated administrator has the following additional privileges:

- Access to all System Settings except activating maintenance mode and rotating the encryption key.

- In the localization app, an elevated administrator can add new languages to OASYS. A normal admin can only edit translations.
- **Users**: Remove or add other users to Admin usergroup
- **Users**: Delete/disable other Administrators
- **Users**: Create Administrators in the Admin usergroup
- **Users**: Modify another Admin level user's account properties
- **Backup**: Download any backup archive file
- **Backup**: Delete any backup archive file
- **Backup**: Restore previous snapshot file

3.2 Backend Layout and Navigation



The main menu is located on the left of the screen. The main menu provides access to the various apps of OASYS.
Depending on the group properties of your user group, the items of the main menu do vary.
This manual covers all the apps that are typically used for OASYS admin tasks.



The toolbar is located on top of the screen. It is context sensitive, meaning its content changes depending on what you are working with. The toolbar provides you with the set of tools that you can currently use.

3.2.1 Admin and Superadmin Apps

For more detailed information, please read the chapters below.

Localization
The purposes of this app are:

- Editing of standard strings displayed in the OASYS frontend, including error messages.
- Adding new test languages to OASYS and providing the translations for the frontend strings.

System Settings
In this app, the variables for configuring OASYS can be used and edited.
You will also find some tools here, such as for system checks.

Users
The OASYS user management.
Here you can edit user groups, access privileges and users.

Backup
Create and restore full or partial backups of the OASYS installation.

Upgrader
Connect to the upgrade server and keep the OASYS installation up to date.

3.2.2 Standard Backend User Apps (Overview)

More detailed information can be found in the user manual.

Folder Permissions

Setting folder-based access permissions is one of the administrative tasks in OASYS.
Apps with folders are:

- Content
- Tests
- Test Takers

For details, see section **Assigning Folder Rights and Assigning Module Access**.

The Dashboard
The dashboard is the homepage of the OASYS backend. From here, you have access to your **account settings** (gear icon), a **logout button**, and **widgets for multiple purposes**.
For more detailed information, please read below.

Locked items

If a backend user sees a padlock icon on one of their bookmarked items, it means they no longer have access to it. This is because the owner has removed or changed the access settings. In order to change the settings, ownership, superadmin or elevated admin privileges are required.

Content
The content manager app is where you can create, edit, and organize the content for tests.

Tests
The Test Manager enables the backend user to create and organize tests or surveys from existing stimuli and / or items.
OASYS 3.6 uses a Draft/Published (Locked) workflow to protect published assessments and their existing results. See the user manual for test-authoring details.

Test Takers
The test-taker manager enables the backend user of OASYS to create and organize test-takers.

Activity Tracker
In the Activity Tracker, every access by test takers to tests is logged and can be reviewed.
A complete log provides a detailed record of behavior during the test.
Authorized users can use **Force logout** to release a stuck test session without resetting its answers, progress, activity, or remaining time. Confirm the affected session before using this action.

Test Results
This app provides access to the test results.
It is also the starting point for manually scoring tests.
The **report builder** is part of the test results menu. The report builder can provide a range of statistics with graphical visualization.
Test Journeys are available in the results and activity views; see the user manual for those workflows.

3.2.3 Content Related Admin Features

The content manager menu has two admin-only features which provide the possibility of transferring page groups (see user manual) from one OASYS installation to another.

Exporting and Importing Page Groups Page groups can be exported from OASYS and imported into the same or another OASYS installation. Within the same installation, this can be useful, for example, if you want to make parts of your content available to another user without having to grant them folder access.

The export consists of a ZIP archive that you can download to your computer. The archive contains the complete page group or all page groups that you have exported together. This includes all pages with all their interactions, as well as all media contained in the page group(s), including unused media.

Please be aware that this feature is not intended for backup purposes. Imports will only be supported within the same OASYS major version. This feature is intended to share content between different OASYS installations. Within the same OASYS installation, we encourage you to use the load existing pages feature.

- **Export page groups**
  This works with multiple selected page groups. The page groups are exported as a single ZIP archive containing all the content. You will be prompted to save the archive on your local computer.
- **Import page groups**
  You will be prompted to select a valid ZIP file to upload and import.
  For details, follow the Export and Import steps below.

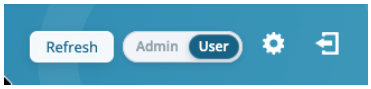
Export

- Select one or more page groups. Use Shift for a range, or Ctrl (Windows/Linux) or Command (macOS) for individual selections.
- Click the **Export page groups** button located in the top bar.
- In case you get an error message, review the page group.
- Save the export file to your local system.

Import

- In the content manager, open the folder you want to import the page group(s) to.
  Make sure that the folder doesn't contain a page group with the same name as one of the page groups you want to import!
- Click the **Import page groups** button located in the top bar.
- The import dialog opens in a pop-up.
- Click the **Choose ZIP package** button.
- Locate the exported page group(s) file you want to import.
- Click the **Review package** button.
- You will get a dialog with an overview of what you are about to import. At this point, you can cancel the import
  In case the file you are trying to import is invalid or too large for the current server settings, you'll get an error message.
- Click **Start import**.

3.2.4 The Dashboard



The dashboard navigation is located in the upper right corner.

- **Refresh** updates all information displayed in the dashboard widgets.
- The **Admin/User** switch toggles the view between admin and standard backend user widgets.
- The **gear button** opens the account settings.
- The **logout button** terminates your session.

System Status

The system status widget provides an overview of key system information and their current status. Warnings and errors are highlighted in color.

Click a status tile to view more detailed information.

Additional checks and system reports are available from the system settings menu.

User Overview

The user overview provides an overview of key user-related information, including the total number of users and user groups, locked accounts, failed login attempts, users with root access, and email configuration. Warnings and errors are highlighted in color.

Click a status tile to view more detailed information.

System Logs

The system logs widget lists all logfiles written by OASYS. Logs highlighted with a colored background contain new entries since your last login.

Click a log file name to open it in the Log Viewer.

Click the corresponding icon to download or clear the file.

Localization

The Localization widget provides an overview of the localization status of the OASYS installation. Changes to the values of keys in the default languages are indicated.

The Additional Languages table lists basic information about additional languages. Clicking a value in the **Missing** column opens a list of untranslated variables.

Chapter 4

User Management

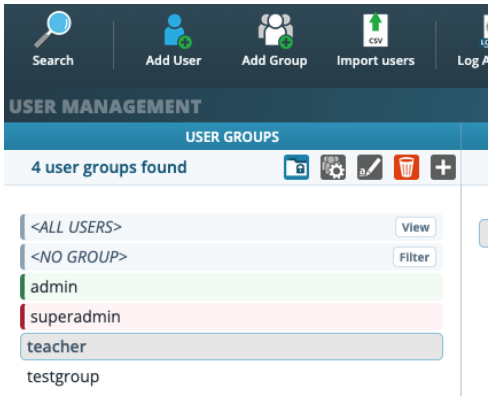
4.1 User Groups

All privileges in OASYS are based on user groups. **There are no individual user privileges, with two exceptions.**

- **Elevated administrator** provides a set of extended privileges.
- **Home folder create** provides the privilege of creating folders at root level. If this is not granted, a user can only create folders within folders their group has write access to.

4.1.1 Creating User Groups

Open the **users app** from the main menu.



Create a user group by:

- Clicking **Add Group** in the top bar.
- Clicking the **plus sign** on top of the user groups column.

You will be prompted for a group name. User group names have to be unique!

Rename a user group by clicking the **rename button** on top of the user groups column.
Delete a user group by clicking the **delete button** on top of the user groups column.

Predefined User Groups and Filters

OASYS has two predefined user groups:

- **admin**
- **superadmin**

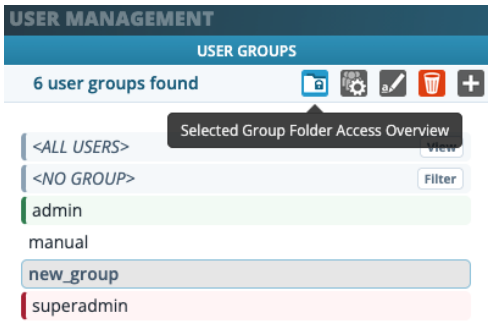
In addition there are two predefined filters:

- **<ALL USERS>** shows all backend users.
- **<NO GROUP>** shows all users who are not assigned to a group.

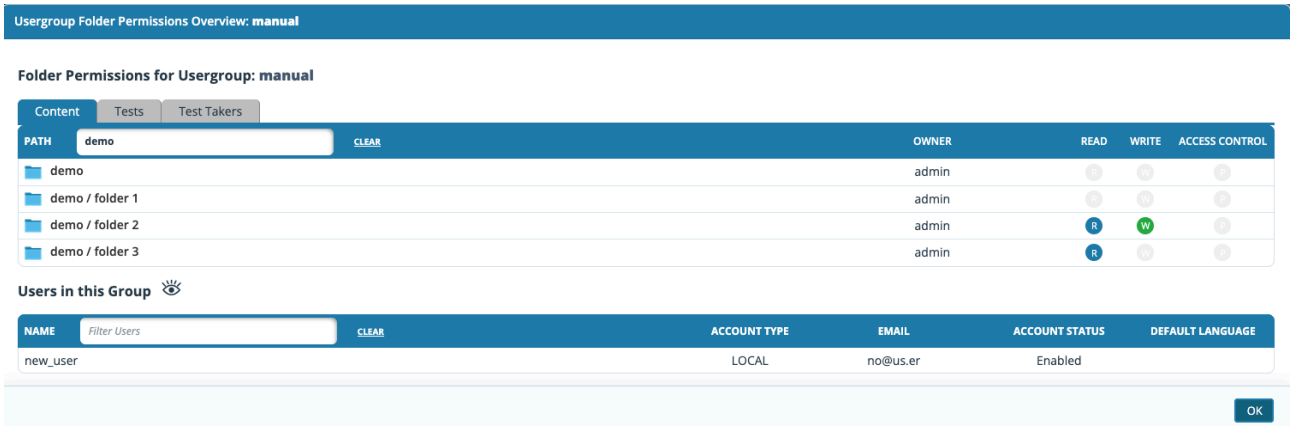
You can't rename or delete the predefined user groups and filters.

4.1.2 Assigning Folder Rights

Folder Permission Overview



Select a user group.
On top of the user groups column, click the access overview button.
The overview is read-only and does not allow editing!



User Group Folder Permission Overview

- You can switch between tabs for content, tests and test takers.
- Folders can be filtered by path.
- The second column indicates the owner.
- The last three columns represent the three different permissions in OASYS:

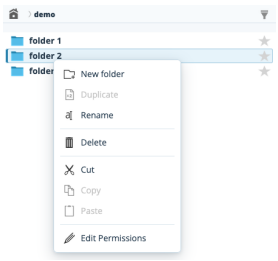
- Read: Members of the current group can view the content of the folder.
- Write: Members of the current group can change the content of the folder.
- Access Control: Members of the current group can change the access settings of the folder.

- **Clicking a folder name brings you directly to the app where you can edit the folder permission.**
- On the bottom, there is an overview of the group's users along with their basic account information.

Assigning Folder Permissions

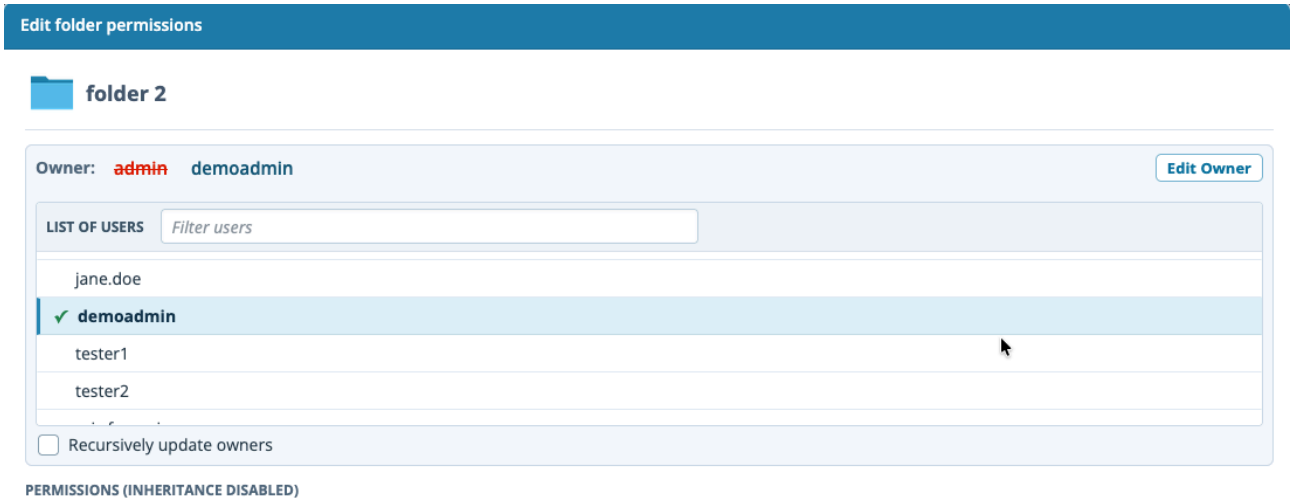
Navigate to the folder you want to assign permissions to. You can do that by:

- Opening the app where the folder is located.
- Clicking the folder in the permission overview.



Right-click the folder.
Click **Edit Permissions** in the context menu.
This opens the Folder Permission Dialog, where you can:

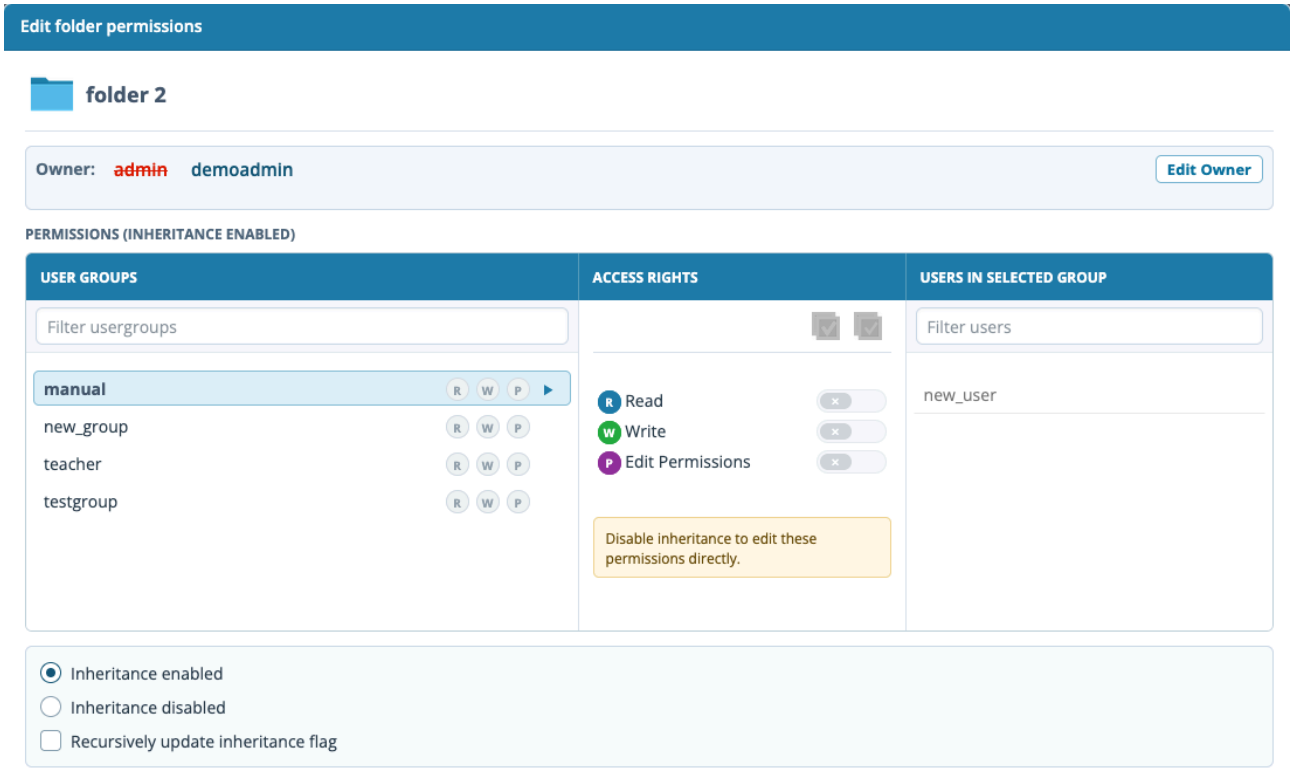
- Change the owner of the folder.
- Edit read, write and permission editing settings of the folder for the different user groups.



Changing the owner

- Click edit owner.
- Select the new owner.
- You can use the search field.

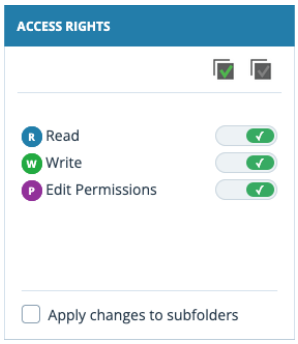
- Check **Recursively update owners** if you want to set the new owner for subfolders, too.



Editing Access Permissions

Select a group to edit its access permissions for the current folder.
Inheritance determines whether you can edit the folder's access permissions directly or if it inherited the permission from the parent folder. If so, you'll have to edit the parent folder's permissions.
You can **disable inheritance** to detach a folder from the inheritance chain.
Recursively update inheritance flag will pass your change of inheritance through the entire tree of subfolders.

Set access permissions by checking/unchecking them.



- Read: Members of the selected group can view the content of the folder.
- Write: Members of the selected group can change the content of the folder.
- Edit Permissions: Members of the selected group can change the access settings of the folder.

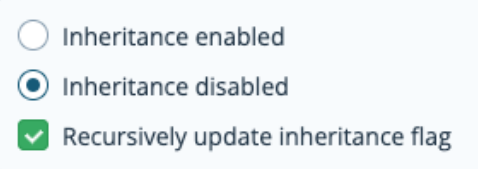
Apply changes to subfolders will pass your changes of permissions through the entire tree of subfolders.

How Inheritance Works

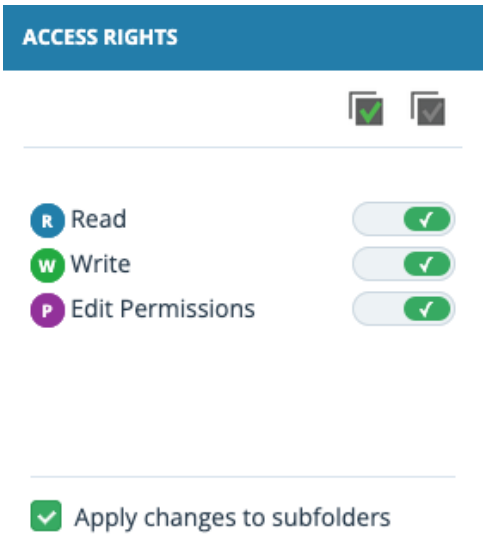
- When you are inside a folder (i.e., not at the root level), the folder can have inheritance enabled. This means it will adopt the permissions of its parent and remain locked to those permissions until inheritance is disabled.
- Users without superadmin privileges can only edit one folder at a time.
- The inheritance switch only works **upward** one level: it copies and locks the permissions from the folder's parent. At non-superadmin level, there is no option to enforce inheritance **downward**. Recursive application of permissions is only available to superadmins.

Supercharged Permission Editing and Bulk Permissions Editing

These features are accessible only to superadmins!



Inheritance flags can be pushed **downward through a tree**, recursively starting anywhere, including the root folder level.
For this, use the settings on the left.

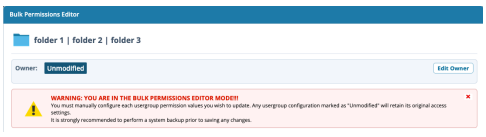


Instead of using **recursive inheritance** to apply and lock permissions across multiple levels, **Apply changes to subfolders allows you to force a permission set downward through a folder tree** on a one-off basis, without setting any inheritance flags.

- First select a folder.
- Select a group.
- Apply the settings and activate the recursive option.

Ownership changes can also be forced down a tree branch recursively, although there is no inheritance option for owners.

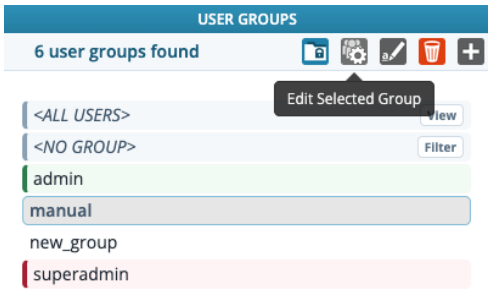
Multiple folders can be selected and modified for mass updates.



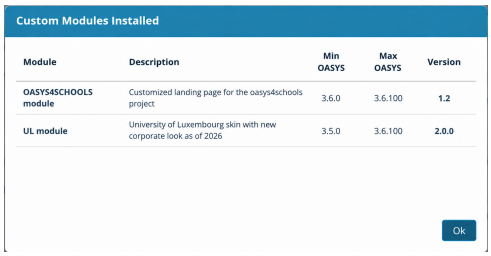
- Select multiple folders using Shift for a range, or Ctrl (Windows/Linux) or Command (macOS) for individual selections.
- Right-click and choose **edit permissions**.
- Choose a group, set the changes and apply them to all selected folders.

4.1.3 Assigning Module Access

Custom modules are functionalities which extend the standard OASYS features. Please see the corresponding section of the manual.



Select a user group.
On top of the user groups column, click the edit selected group button.

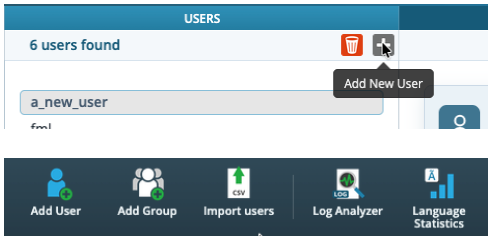


By default, no modules are activated.
Select the modules you want to give the group access to and confirm with Ok.

4.2 User Creation and Authentication

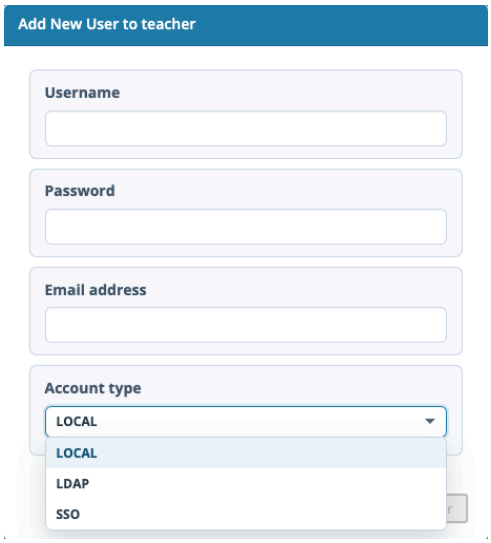
4.2.1 Creating Users

Open the **user app**.
As a first step, you have to **select the group you want the user to be created in.**



Create a new user by:

- Clicking the **plus button** on top of the users column
- or clicking the **Add user button** in the top bar.



- **Username:** The username must be unique.
- **Password:** The password field is only available for local account types!
  Users can change the password in their settings.
- **Email address:** The forgot my username or password link on the login screen requires a valid email address. (**Email functionality must be configured in System Settings.**)

- **Account Type:** The type of authentication used for the login.
  - **LOCAL:** A password hash is stored in OASYS.
  - **LDAP:** LDAP (Lightweight Directory Access Protocol) is a protocol used to access and manage directory services, such as user and group information, over a network. It is commonly used for authentication and storing organizational data.
  - **SSO:** Single Sign-On (SSO) allows users to log in once and automatically access all connected applications without needing to sign in again.

All account types other than LOCAL require setup in the system settings!

The No Group Filter

Users are created within a group. Deleting a group does not delete its users, who may belong to other groups. Users left without a group can be located with the No Group filter.
Use the No Group filter to assign a new group to these users or delete accounts that are no longer needed.

4.2.2 Authentication Methods

Local Authentication

The username and a password hash are stored in the OASYS database. This authentication method doesn't require any additional setup.

LDAP Authentication

This authentication method requires an LDAP server (e.g., Microsoft Active Directory, OpenLDAP)
Navigate to the **system settings**.
Change the following settings according to the parameters of your LDAP server:

- **ldap_appUser:**
  The username used by the application to connect to the LDAP server and perform user searches. This is often a service account with read permissions.
- **ldap_appPass:**
  The (encrypted) password for the ldap_appUser. Needed to authenticate the search account on the LDAP server.
- **ldap_server:**
  The address of the LDAP server. It must start with ldap:// (optionally upgraded with StartTLS) or ldaps:// (implicit TLS). Example: ldaps://ldap.example.com.
- **ldap_searchBase:**
  The starting point in the LDAP directory tree for user searches, written as a Distinguished Name (DN). Example: ou=users,dc=example,dc=com.
- **ldap_query:**
  The search pattern used to find users. %1 is a placeholder that will be replaced by the username entered at login. Example: cn=%1 searches by common name.
- **reset_LDAP_Redirect:**
  This is the URL where users will be redirected if they need to reset or retrieve their LDAP password information.
  The URL must start with http:// or https://.
  It usually points to an external password management or support page (since LDAP passwords typically can't be reset directly through the application).
- **ldap_caCertificateFile**
  Optional absolute path to a PEM CA certificate bundle used to verify the LDAP server certificate (the system trust store is used when empty)
- **ldap_networkTimeoutSeconds**
  LDAP connection and network timeout in seconds.
- **ldap_searchTimeoutSeconds**
  LDAP user search timeout in seconds
- **ldap_startTls**
  Upgrade LDAP:// connections to TLS before binding (recommended when the LDAP server supports StartTLS)
- **ldap_stripLoginDomain**
  Remove the @domain suffix from usernames before searching LDAP

- **ldap_verifyCertificate**
  Verify the LDAP server TLS certificate against the configured CA file or system trust store (recommended)

Single Sign-On (SSO)

OASYS 3.6 supports SAML 2.0 for backend authentication. Coordinate setup with the identity-provider administrator.

1. Copy conf/ssodata.default to conf/ssodata.ini and configure the service-provider issuer, identity-provider endpoints and issuer, user identifier attribute, and signature algorithm to match your federation.
2. Check the externally reachable ACS URL, normally https://your-domain/path-to-oasys/editor/sso.php.
3. Register the service provider with the identity provider using the metadata at editor/tools/sp_metadata_writer.php under your OASYS URL.
4. Configure HTTPS and set cookieSecure=true and cookieSameSite=None for SAML authentication.
5. Enable SSOSysActive in System Settings and test with an appropriately configured SSO account. Keep a working local superadmin account for recovery.

Consult the separate OASYS SAML configuration guide for the configuration fields and federation-specific details. Do not edit obsolete inc/php/ssodata.php or template paths.

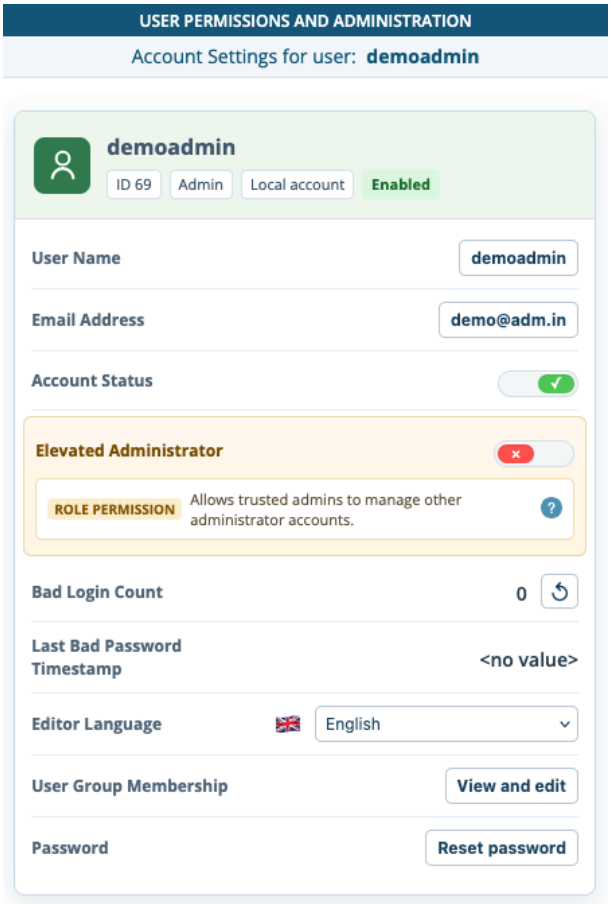
4.2.3 User Options and Settings

User Groups and Privileges

All privileges in OASYS are based on user groups. **There are no individual user privileges, with two exceptions.**

- **Elevated administrator** provides a set of extended privileges.
  This option is only available for members of the **admin user group**.
- **Home folder create** provides the privilege of creating folders at root level. If this is not granted, a user can only create folders within folders their group has write access to.

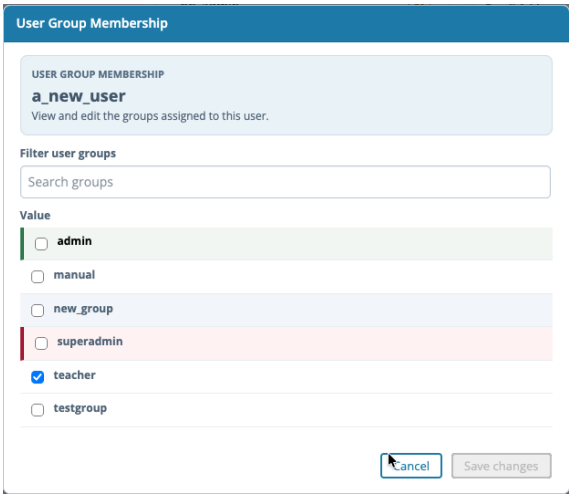
If you want to grant access or administrative privileges to a user, you have to make them a member of the corresponding group. **A user can be a member of multiple groups!**



Open the **user app**.
Select one of the groups the user you want to edit is in, then select the user. Alternatively, you can use the Search button in the top menu.
In the right column, you find the settings for the user.
Click a value to edit it.

- **User Name:**
  The user name can be changed. As the name has to be unique, OASYS won't accept a name, which is already in use.
- **Email Address:**
  User email. Users can change their email address in their user profile.
- **Account Status:**
  Accounts can be manually disabled or enabled.
  After 20 failed login attempts, a user will be automatically disabled.

- **Elevated Administrator:**
  Only available for users who are members of the admin group.
  The elevated admin is an admin with extended privileges. See chapter Admin Roles and Permissions for details.
- **Home Folder Create**
  Privilege to create folders at root level.
  This setting is only available for users who are not members of the admin or superadmin group.
- **Bad Login Count:**
  The count of failed login attempts. This is read only.
- **Last Bad Password Timestamp:**
  Time of the last failed login.
- **Editor Language:**
  Language of the OASYS backend. Users can change this setting in their profile.
- **User Group Membership:**



The user will be assigned to all checked groups.

- **Password:**
  Users can change their password in their user profile.
  If a user is locked out from their account, the password can be reset with this option.

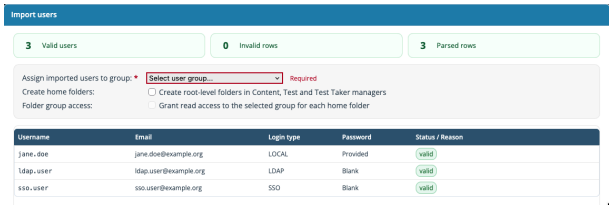
4.3 Import Users

OASYS allows you to import backend users from a CSV file into a regular user group; importing into admin or superadmin is not allowed. LOCAL users require a password; passwords must be blank for LDAP and SSO users. Resolve duplicate usernames, duplicate email addresses and validation errors before confirming. Handle import files containing credentials or personal data confidentially. To begin, click the **Import users** button in the user manager's top bar.

oasys_users_import_example

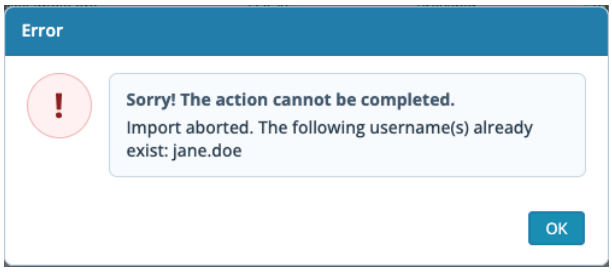
username	email	login_type	password
jane.doe	jane.doe@example.org	LOCAL	ChangeMe123
ldap.user	ldap.user@example.org	LDAP	
sso.user	sso.user@example.org	SSO	

The structure of the CSV file is simple.
Just specify **username, email, login_type** (see above) and, in case of login_type LOCAL, a **password.**



First, the uploaded CSV file will be parsed.

- Specify the group the users will be initially assigned to.
- Optionally, you can activate **Create home folders**. This option will create a folder named with the user name in the Content, Tests and Test Takers managers. The created user will be its owner.
- Additionally, you can grant read access to the home folders for the group you specified above.



In case of an error, you'll get a message indicating the problem. Please fix the CSV and retry.

4.4 Logs Analyzer

The logs analyzer records a specific set of backend user activities, along with the date of the action and the session ID.
The purpose of the Logs Analyzer is to make deletions or renaming of objects by backend users traceable. This is especially important when users report that content they were working on can no longer be found.
The logged **modules** are:

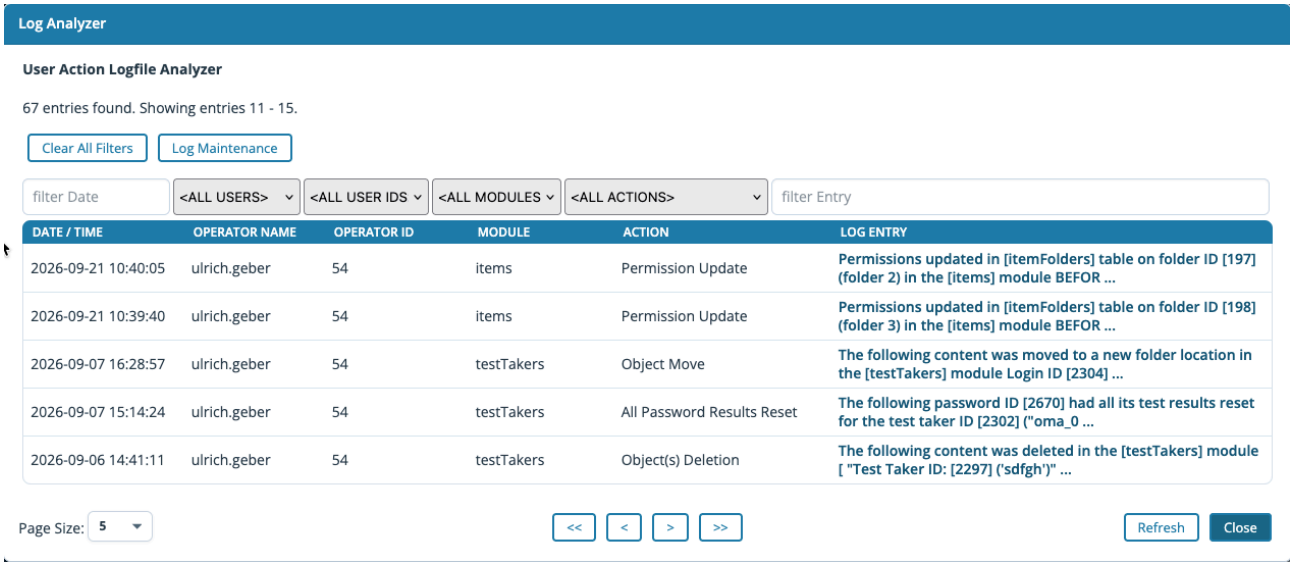
- Contents (items)
- Tests
- Test Takers

The logged **actions** are:

- Renaming of objects
- Permission update on folders
- Folder owner change
- Object(s) deletion
- Item deletion
- Object move to a different folder
- Test taker results reset
- All password results reset
- Test/password results reset
- Test results reset

If an action is greyed out in the dropdown, it is not present in the current list of entries. (According to the filters applied)

4.4.1 Monitoring User Activities



The Logs Analyzer

Columns

- **Date/Time:**
  The timestamp of the action

- **Operator Name:**
  The backend user's login name.
- **Operator ID:**
  The backend user's database ID
- **Module:**
  The module where the action was taken.
- **Action:**
  The type of the action
- **Log Entry:**
  The corresponding log entry, along with the session ID.
  Click it for the full text version.

Filters

- **Date:**
  Narrows the timeframe of the entries to a specific date.
- **Users:**
  Select a specific user from the dropdown.
  Only users having corresponding log entries are shown.
- **User IDs:**
  Same as users, but listing the users' database IDs.
- **Modules:**
  Narrows the list to a specific module.
- **Actions:**
  Narrows the list to a specific action (see above for the list of logged actions).
- **Filter Entry:**
  Search for entries containing the given search string in the text.

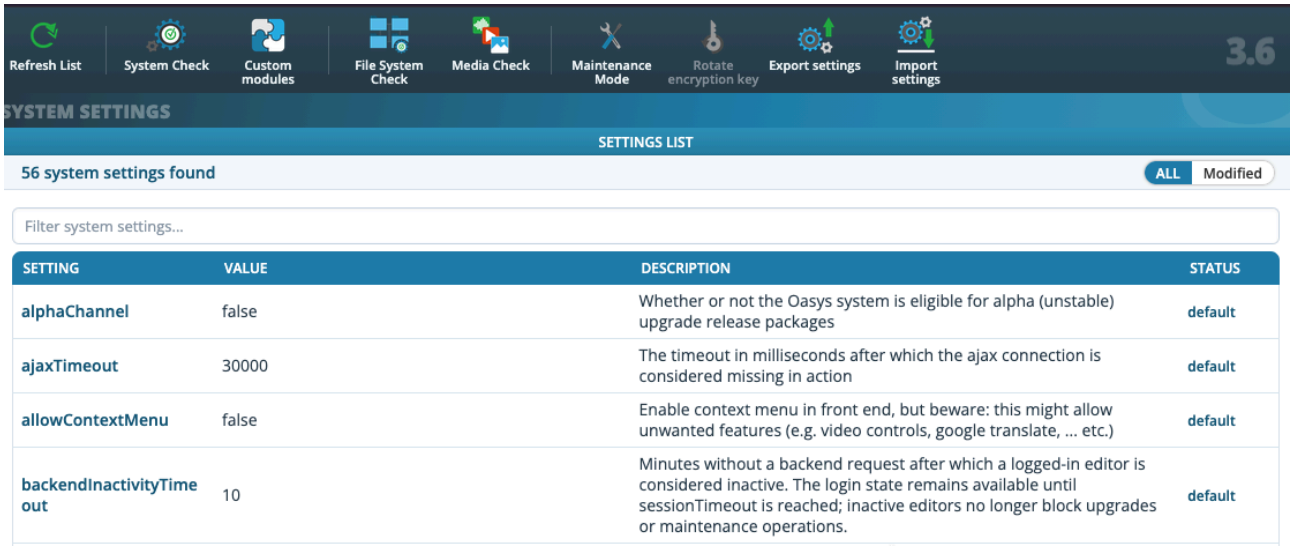
Clear all Filters:
Removes all currently active filters from the list.
Log Maintenance:
This feature requires superadmin privileges!
Remove older entries from the list according to the time limit chosen from the dropdown.

Chapter 5

System Configuration

5.1 Configuring OASYS

OASYS is configured via the **system settings app (main menu)**. All settings are managed through system variables. An exception is the SSO configuration, which requires editing files. Another tool for configuration is the localization app. It can be used to add additional test languages to an OASYS installation.



System Settings

Use the toggle switch to show all or just the modified settings.

- 1. Column: Name of the system variable
- 2. Column: Current value
- 3. Column: Description
- 4. Column: Modification info

In order to modify a value, click the first or the last column of the entry.

5.2 System Variables

- **alphaChannel**
  This is an upgrader setting. If it is set to false, the upgrader will only fetch stable releases. If true, the upgrader will fetch prereleases as well. This is only recommended if you need a feature from a prerelease before the next stable release is published.

- **ajaxTimeout**
  Defines the time limit (in milliseconds) for how long the system waits for an AJAX response before considering it failed. Adjust this according to your server's responsiveness and performance.

- **allowContextMenu**
  Enables or disables the browser's context menu (right-click). Activating it may allow unintended features like video controls or Google Translate.

- **backendInactivityTimeout**
  Specifies how many minutes an editor can go without sending a request to the backend before being considered inactive.

  The editor remains logged in until the sessionTimeout is reached. However, inactive editors no longer prevent upgrades or maintenance operations.

- **cookieSameSite**
  Specifies the SameSite policy for the frontend and backend state cookies.

  The None setting is required for SAML authentication and requires secure cookies.

- **cookieSecure**
  Specifies whether frontend and backend state cookies are sent only over secure HTTPS connections.

- **customLoginURL**
  By default, a test taker will be redirected to the login screen, after they have finished a test. Use this setting to override this behaviour and set a custom URL instead.

- **debugSystem**
  If set to true, detailed error messages for debugging purposes will be displayed in the browser.
  Warning: Should not be enabled in production environments.

- **developmentMode**
  When enabled, the system loads CSS and JS files directly (not from blobs).
  Warning: Not for use in production.

- **forceLogoff**
  Automatically logs users off when a test becomes inactive due to time or date restrictions. This can be overridden at test level.

- **frontendFailedAttemptWindow**
  Specifies the time window, in seconds, during which failed frontend login attempts are counted.

- **frontendLoginLockoutSeconds**
  Specifies how long a frontend login remains locked, in seconds, after too many failed login attempts.

- **frontendMaxFailedAttempts**
  Specifies the maximum number of failed login attempts allowed per frontend login during the configured time window.

- **hideTimeoutMsg**
  Controls whether timeout messages are shown in new tests. If true, the timeout message is hidden.
  This can be overridden at test level.
- **landingPage**
  The default frontend login page can be replaced with this setting.
  Select a landing page available in the installation's landingPages directory.
- **ldap_appUser**
  Username for the LDAP search account.
- **ldap_appPass**
  Encrypted password for the LDAP search account.
- **ldap_caCertificateFile**
  Optional absolute path to a PEM-encoded CA certificate bundle used to verify the LDAP server's certificate. If left empty, the system trust store is used.
- **ldap_networkTimeoutSeconds**
  Specifies the LDAP connection and network timeout in seconds.
- **ldap_server**
  URL of the LDAP server (must start with LDAP:// or LDAPS://).
- **ldap_searchBase**
  Distinguished Name (DN) for the user search base.
- **ldap_query**
  Query string for searching users; %1 is replaced by the username.
- **ldap_searchTimeoutSeconds**
  Specifies the LDAP user search timeout in seconds.
- **ldap_startTls**
  Upgrade LDAP:// connections to TLS before binding (recommended when the LDAP server supports StartTLS).
- **ldap_stripLoginDomain**
  Remove the @domain suffix from usernames before searching LDAP.
- **ldap_verifyCertificate**
  Verify the LDAP server TLS certificate against the configured CA file or system trust store (recommended).
- **limitNavigation**
  Default setting for limiting navigation during tests.
  This can be overridden at test level.
- **loginLanguage**
  Default language activated at login.
  Note that all buttons for choosing a language at login will still be available. This setting just controls the preselection.
- **logoAdminPanel**
  The logo used in the admin panel can be replaced.
  The file must be located within the OASYS installation folder. The path must start without a /.

- **logoLoginView**
  The logo used in the frontend login screen can be replaced.
  The file must be located within the OASYS installation folder. The path must start without a /.

- **logoBackendLoginView**
  The logo used in the backend login screen can be replaced.
  The file must be located within the OASYS installation folder. The path must start without a /.

- **logPayload**
  Indicates how much of the frontend network traffic is logged (nothing, just packages or packages with sent data).
  Log entries are stored in the database. Table: logTransfers

- **mediaLocation**
  Location where media is stored (on disk or in the database); **changes require OASYS team support**.

- **menuLanguages**
  List of languages displayed on the frontend login screen.
  If you deactivate all but one language, the language selection buttons will not be displayed.

- **mutationMethod**
  Default mutation method setting for new mutation tests.
  Method can be set to random (a random test will be picked) or to sequential (the next test from the list will be picked).
  This can be overridden at test level.

- **optimiseDataTransfer**
  Reduces data transfer by discarding queued answer events when the same field is changed multiple times. This may result in an incomplete history of the test taker's interactions.

- **passwordField**
  Defines whether the password field in the frontend login displays the typed text or hides it.

- **retryCount**
  Number of attempts to send test data to the server before OASYS considers the connection as failed.
  If the number is exceeded, the test will be blocked. This prevents test takers from continuing with a test which can't send test results to OASYS.

- **saveResults**
  Default setting for whether test results are saved when creating a new test.
  This can be overridden at test level.

- **sendFrequency**
  Interval (in seconds) between sending data packages (test results) from the frontend to the server.

- **sessionTimeout**
  Time (in minutes) to retain a login session.

- **showScore**
  Default setting for displaying scores in new tests.
  This can be overridden at test level.

- **title**
  The title displayed in the browser tab (for the OASYS frontend). Also used by iOS when creating a homescreen link.

- **useTimer**
  Default setting for enabling the ”use timer” option in new tests.
  This can be overridden at test level.

- **waitForMediaCache**
  Default setting for media caching in new tests. If set to true, the test will start when all media is cached.
  This can be overridden at test level. (Name of the property: Wait for media to load)

- **writeLog**
  Determines whether technical log data is written. When enabled, OASYS writes a multitude of logs, for backend as well as for frontend user actions.
  The logs are mostly for the OASYS team. If support is needed, we might request some specific logfiles from your instance.
  Logs are located in the logs folder at the root of your OASYS installation.

- **SSOSysActive**
  Indicates whether Single Sign-On (SSO) integration is active.

- **emailSysActive**
  Indicates whether email functionality is enabled for automated account resets, account validation, etc.

- **SMTP_Host**
  Outgoing email server address for OASYS Editor.

- **SMTP_Port**
  Email server SMTP port used by OASYS Editor.

- **SMTP_Username**
  Email server SMTP username for OASYS Editor.

- **SMTP_Password**
  (Encrypted) Email server SMTP password for OASYS Editor.

- **SMTP_From**
  The ”FROM” address used for outgoing emails.

- **SMTP_Encryption**
  Defines the encryption type for the SMTP server connection (none, implicit, or TLS upgradable).

- **reset_LDAP_Redirect**
  URL for redirecting LDAP password reset information (must start with ”http(s)://”).

- **upgraderURL**
  The URL the upgrader app and the upgrader notification use. Don’t change this unless the OASYS team does notify you of a server change!

5.3 System and File System Check

In order to perform a system check or a file system check, navigate to the system settings in the main menu. The buttons for both checks are located in the top bar.

5.3.1 System Check

The system check verifies whether the server environment meets the technical requirements to properly run OASYS.
In summary, this check is a quick way to validate that the hosting environment is correctly configured and compatible with the currently installed OASYS version.
It checks:

- PHP version, required PHP modules and configuration
- Database system and version
- Server and database configuration limits (e.g., memory, execution time)
- User privileges
- System time synchronization

The purpose of the system check is:

- To ensure compatibility between OASYS and the server.
- To prevent runtime errors due to missing modules or incorrect configurations.
- To confirm that the database setup and user permissions meet the application's operational needs.

When to perform this check:

- After updates to the server, PHP, database, or the application itself.
- When migrating OASYS to a new server.
- During troubleshooting if OASYS is not working as expected.

5.3.2 File System Check

The file system check verifies the integrity of OASYS' file structure by comparing the current files against a baseline (a known, correct file set).
It looks for:

- Additional files (unexpected files that might indicate tampering or leftover files)
- Missing files (required files that are no longer present)
- Modified files (files that differ from the original version, based on checksum)

The purpose of the file system check is:

- To detect unauthorized changes or file corruption.

- To ensure that no critical files have been accidentally deleted or altered.
- To identify potential security risks from extra or modified files.

When to perform this check:

- Regularly as part of system maintenance.
- After a software update or patch.
- If there is suspicion of a security breach.
- After manual file system changes or migrations.

Clean up Files:
If there are leftover files from old versions of OASYS which are not part of the exclusion list, the button will remove those files permanently. This is to keep the system in line with the exact system files that should be there. If a file is found to be ”different” by not matching the checksum value, that file is **not** removed with the Clean up Files button.
Only files which are

- not in the checksum list and
- not in the exclusion list

are removed with this cleanup button.

5.3.3 Media Check

This check scans all media and custom-content files and may take several minutes on a large production system.

Run the process during a period of low system activity to minimise competition with ongoing assessments for disk and database resources.

The check is performed on media used in special pages belonging to tests (legal, score screen...) as well as on media references found on test pages.

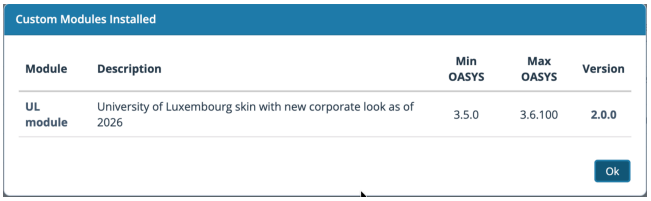
If issues are found, review the list and take a current backup before choosing **Fix issues**. This action can move or delete files and modify database entries; review the results afterwards.

For test-content media it may move or delete orphaned files, remove stale database entries, and synchronize disk/database storage. For meta-page media it copies cross-test references into the correct test folder, updates metadata, removes empty folders, and repairs folder writability.

Once the operations are complete, a log is displayed along with the results of a post-check.

5.4 Custom Modules

Custom Modules are modules that can extend, modify, or partially replace functionalities of OASYS. This could include, for example, advanced test management features or custom skins.
If you plan to use OASYS and require extensions, feel free to contact us through the OASYS website.
Custom modules are typically integrated via the main menu or are available as additional options within specific apps.
The **custom modules button** in the top menu of the **system settings** provides an overview of the installed custom modules:



- Min OASYS: Minimum required OASYS version for the installed version of the module.
- Max OASYS: Maximum tested OASYS version for the installed version of the module.
- Version: Currently installed version of the module.

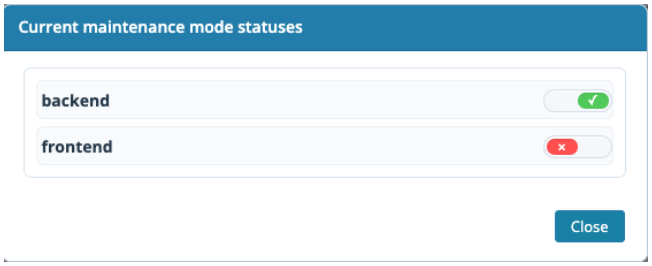
Custom modules are usually delivered with a custom OASYS installer.
The OASYS **upgrader** will recognize the installed module and integrate its newest version in the upgrade process.
In case there is no compatible version of the module, the upgrader will issue a warning and will not perform the upgrade.

5.5 Maintenance Mode

Maintenance mode is a system state that temporarily locks the web application for regular users while allowing administrators to carry out system updates, configurations, or troubleshooting tasks without interference.
Purpose:

- Protects system integrity during updates.
- Prevents user disruptions and potential data inconsistencies.
- Provides a safe environment for backend maintenance.

Maintenance mode is typically used during planned updates, configuration changes, or system checks that could temporarily impact normal operation.
OASYS has two separate maintenance modes:



- **Backend:**
  Only the backend is set to maintenance mode. All backend users are locked out, except superadmins.
- **Frontend:**
  The interface for test takers is locked. The backend is accessible for all backend users.

5.6 Rotate Encryption Key

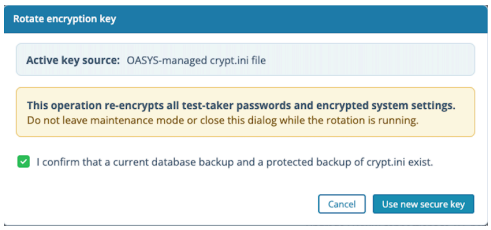
OASYS uses encryption to protect test-taker passwords and sensitive system settings; backend LOCAL account passwords are hashed. Only superadmins can rotate the encryption key.

The Rotate Encryption Key function allows you to replace the current encryption key used to protect stored passwords.

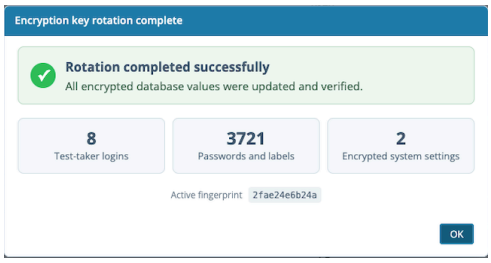
The procedure depends on how your OASYS installation is configured: the encryption key may be managed through a crypt.ini file or through server variables. Follow the instructions for the configuration method used by your installation.

5.6.1 Crypt.ini

Enable both frontend and backend maintenance mode before proceeding. Back up the database and securely preserve its matching key and IV first.



Before rotating the encryption key, make sure that a current database backup and a securely stored backup of the crypt.ini file are available. Confirm this by selecting the checkbox.
The operation re-encrypts all test-taker passwords and encrypted system settings using the new key. Do not leave maintenance mode or close this dialog while the rotation is in progress.
Click **Use new secure key** to start the operation.

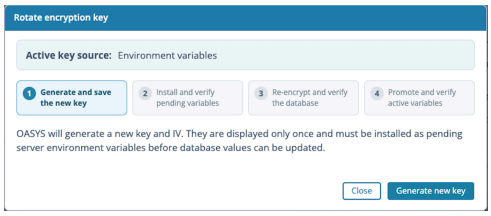


When encryption-key rotation completes successfully, all encrypted database values have been updated and verified. Back up the new conf/crypt.ini together with a matching database backup. Retain the old key material with older backups.
The summary shows the number of affected test-taker logins, passwords and labels, and encrypted system settings. The active fingerprint identifies the new encryption key and can be used to verify which key is currently active.
Click **OK** to close the dialog.

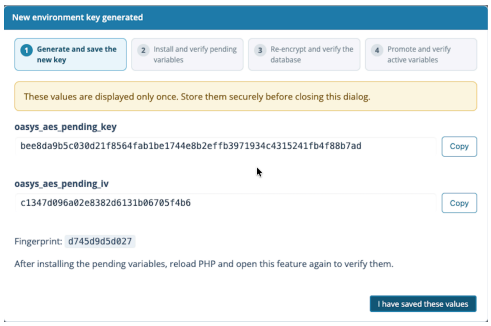
5.6.2 Server Variables

Enable both frontend and backend maintenance mode before proceeding. Back up the database and securely preserve its matching key and IV first.

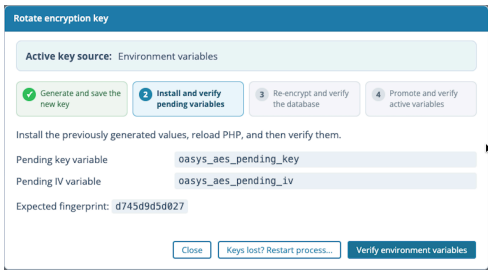
OASYS will lead you step by step through the process.



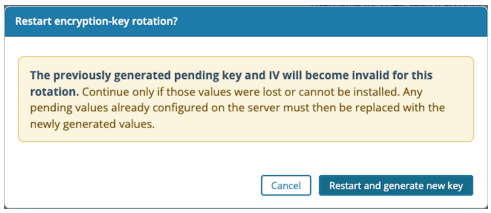
Choose **Generate new key**. Keep the current active key and IV unchanged until the dialog requests promotion.



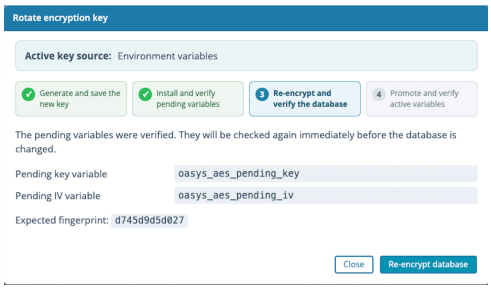
Securely record the generated key and IV immediately; they are displayed only once. Install them as the pending environment variables using the exact names shown in the dialog, while retaining the current active variables.



Reload or restart the service that supplies PHP's environment (for example PHP-FPM or Apache, depending on the deployment). A terminal-only variable change may not reach the web application. Choose **Verify environment variables** and continue only after verification succeeds.



This is an optional recovery action before re-encryption. Use **Keys lost? Restart process...** only if the generated pending values were lost or cannot be installed. Restarting invalidates those pending values; install the newly generated values instead. Do not treat this dialog as a normal rotation step.



After pending-variable verification succeeds, choose **Re-encrypt database**. Keep both sets of variables available and both maintenance modes enabled. Wait for completion before changing the active values.



When prompted, replace the active key and IV environment-variable values with the verified pending values. Reload the PHP service again, then choose **Verify active variables and finish**. Follow the completion instructions before removing pending variables. Verify encrypted settings and test-taker credentials, create a fresh backup with matching key material, and then disable maintenance mode.

Key Rotation and Recovery

Never replace the active key and IV before re-encryption is complete and the tool requests promotion. Retain the old key material with older database backups. If rotation is interrupted, keep maintenance enabled and both sets of variables available. Reopen the tool and follow its reported stage; use **Resume database rotation** if offered. Contact OASYS support if the state is unclear. Restore a database only with the matching key material.

5.7 Export Settings and Import Settings

OASYS allows you to export modified system settings to a CSV file, which can then be imported into another OASYS installation or after a reinstallation.

This makes it possible to transfer specific configurations, such as LDAP settings, without having to configure them again manually.

Export:

Before the export file is created, a list of all included settings is shown. All settings with OASYS default values will be omitted. The CSV uses columns setting and value. Encrypted settings are exported in clear text when the file is created: protect the CSV and restrict access to it. A settings export is not a database or encryption-key backup.

Import:

Select the exported (or created/modified) CSV file.

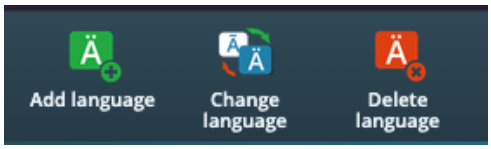
The settings to be applied are displayed before any changes are made.

If everything is as intended and without errors, click **Apply settings**.

Chapter 6

Localization

OASYS is natively designed for multilingual tests. The approach is to provide each content element in multiple languages simultaneously during creation.



When working on a test page, you can add languages, remove languages or change the current language.

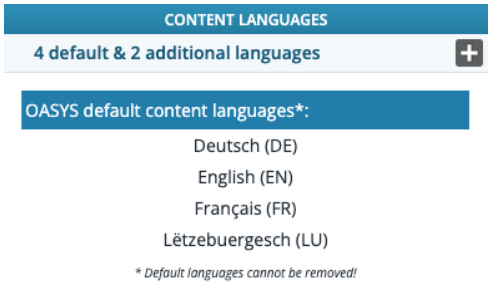


Within the test page, you can switch languages using tabs.

The language of the menu elements in the frontend will change according to the language currently selected by the test taker.
The available languages are defined in the **localization app**.

6.1 Adding New Languages

Adding languages to OASYS requires superadmin or elevated admin privileges!
Languages are added via the **localization app (main menu)**.

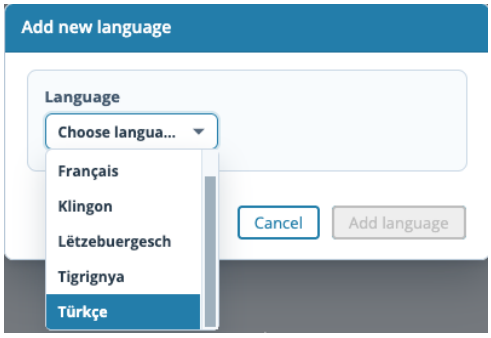


In the right column, there is the list of the default languages.
If there are additional languages, they are listed below.
Click the **plus sign** on top of the column to add a new test language.



Enter

- The **language shortcode** which is used wherever the name is abbreviated.
- The **language name** in the respective language.
- A **fallback language** which will be used where a translation is missing.



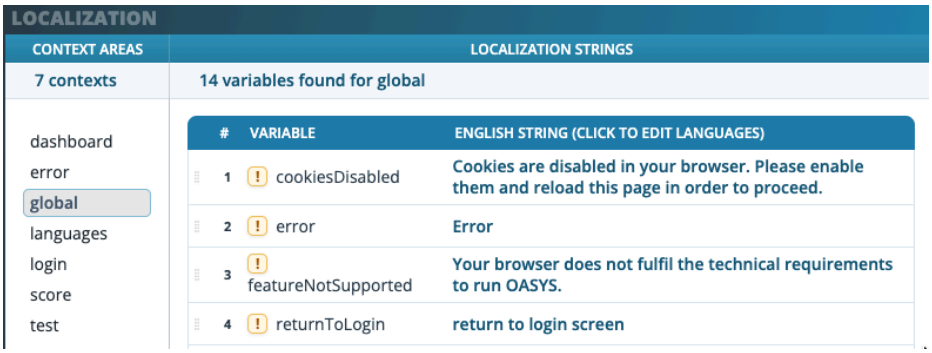
The additional language will be available when **adding a language to a content page.**



And it will be available for **activation for a test.**

6.2 Providing Translations

When adding a new language to OASYS, translations for skin elements (buttons, menu elements) and messages must be provided.
OASYS stores these translations as string values in variables. If a string value is not available in a language, the value of the fallback language will be used.
String translation is done in the **first two columns of the localization app**.

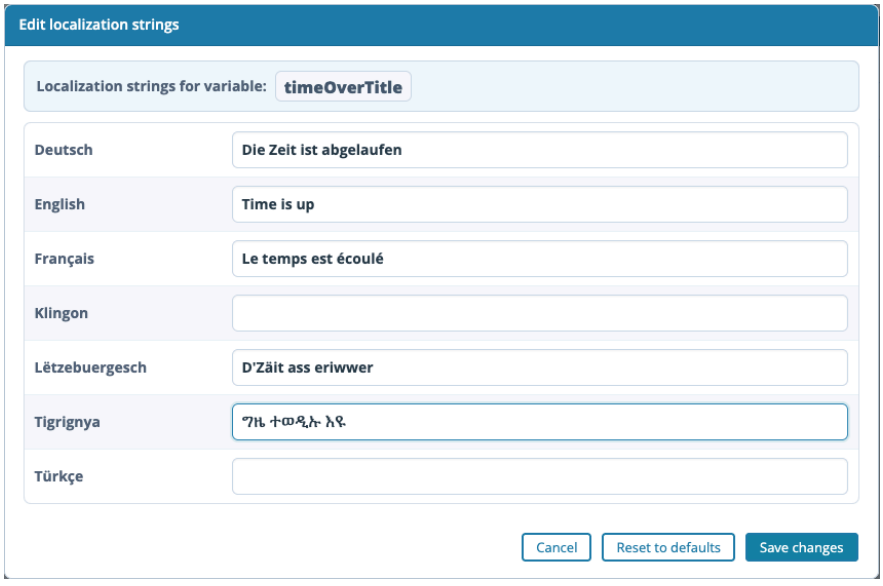


Context Areas

Context areas are groups of variables containing strings related to the same topic.
Select a context area to show its variables in middle column.

Localization Strings

The second column lists all variables of the selected context.
The **variable** column shows the name, the other column shows the current English string value of the variable.
Click the string value to edit the translations.



The translation dialog lists all currently available test languages. You can:

- edit an existing translation.
- add missing translations

Once all variables of all contexts are translated, the localization is complete.

Chapter 7

Backup and Restore

OASYS has two different backup options.
Both are located in the **backup app in the main menu.**
A **snapshot** creates a backup of the *database* and all *uploaded media* at the current point in time.

Use when:

- You want to quickly save the current data state before making minor changes (e.g., content updates, user imports).
- A fast, lightweight backup is sufficient.

A **full backup** includes the entire *OASYS data and file system*, excluding system logs and operational temporary files.

Use when:

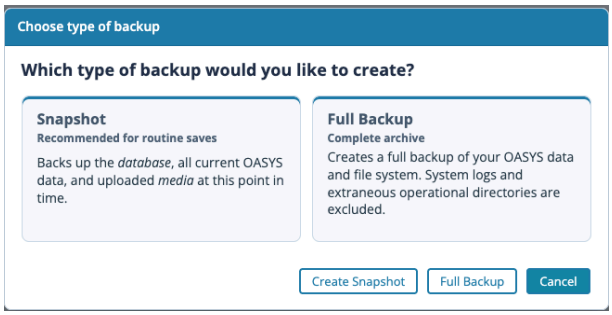
- Performing system upgrades or major configuration changes.
- Migrating the application to another server.
- Creating a periodic comprehensive backup for disaster recovery.

Keep protected copies of backups off the application server. Preserve the key and IV matching each backup, including snapshots: a snapshot does not include conf/crypt.ini or external environment variables. A snapshot made before key rotation requires its matching older key material. Test recovery on an isolated instance before relying on a backup.

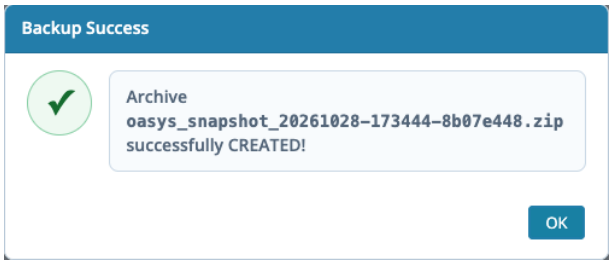
7.1 Database Snapshots

7.1.1 Creating a Backup

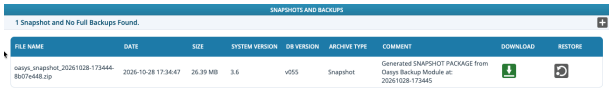
Open the **backup** app.
Click the **plus sign** in the upper right of the backup list.



Choose **Create Snapshot**.



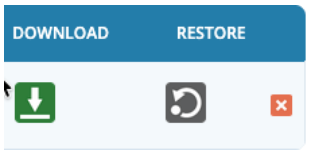
You'll get a success message.



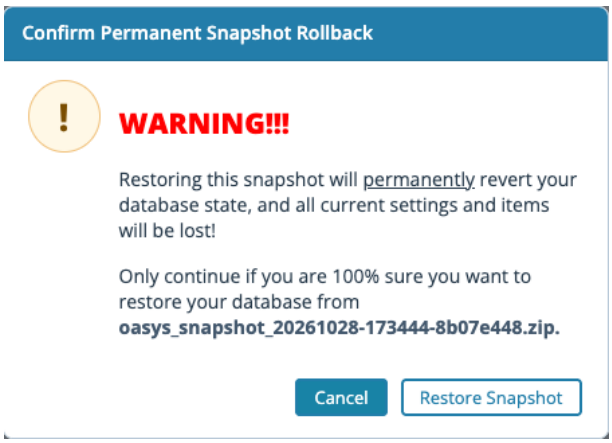
The snapshot will appear in the backup list.

7.1.2 Restoring from Backup

Open the **backup** app.
Locate the snapshot you want to restore in the backup list.



Click the **restore button** next to your snapshot.



Read the warning before you continue.

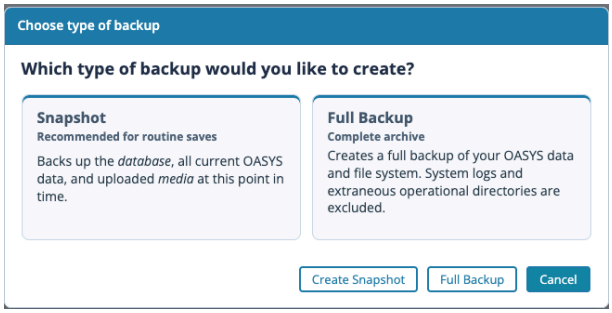
Snapshot Compatibility

An older database snapshot might not be compatible with a newer OASYS version!
OASYS has a built-in compatibility check for this scenario.
An old snapshot will be blocked from being restored when its database version is not compatible with the current OASYS version.

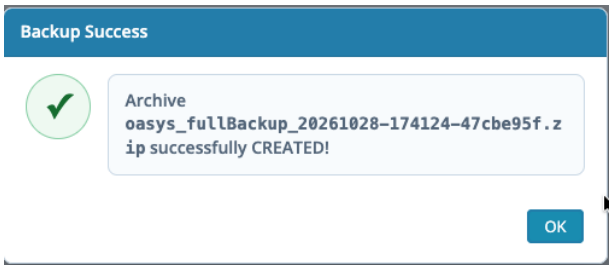
7.2 Full System Backups

7.2.1 Creating Full Backups

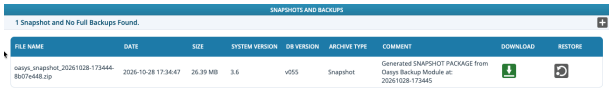
Open the **backup** app.
Click the **plus sign** in the upper right of the backup list.



Choose **Full Backup**.



You'll get a success message.



The backup will appear in the backup list.

7.2.2 Restoring Full Backups

A full restore replaces the current files and database. Schedule downtime, block frontend and backend access, and take a separate recovery copy of the current files, database and encryption configuration before proceeding. Confirm the target instance and backup version. Prefer testing the archive on an isolated instance first.

Preserve the Encryption Keys Before Restoring

The restored database must be paired with the key and IV that encrypted it when the backup was created. A newly generated key, or the current key after a later rotation, is not a substitute. Without matching key material, encrypted test-taker passwords and system settings cannot be recovered correctly.

- **File-managed keys:** The OASYS 3.6 full-backup routine includes conf/crypt.ini and conf/env.ini when these files exist and are successfully archived. Before deleting the current installation, inspect the ZIP and confirm that the expected files are present. Keep a separate protected copy of the backup's matching key material. Restore crypt.ini together with its database; do not allow OASYS to generate a replacement key.
- **Environment-managed keys:** Server environment variables are not captured by the application backup. conf/env.ini maps variable names; it does not contain the environment's key values. Securely retain the actual key and IV used at backup time, their variable names and the deployment configuration needed to supply them to PHP. Restore those matching values on the target server and reload or restart the PHP service before allowing OASYS to access the restored database.
- **Check which source is active:** Valid environment-provided keys take precedence over crypt.ini. Verify the target PHP environment even when the archive contains that file. Do not assume a restored file overrides environment values left over from a newer installation.
- **Backups from before key rotation:** Retain their matching old key material for as long as the backups are retained. Test restoration in an isolated environment. If the required keys are missing, stop before replacing the current installation and resolve recovery with OASYS support.

Treat the archive and key copies as confidential: a full backup may contain both encrypted data and its decryption key. Keep the instance inaccessible during restoration, verify encrypted settings and test-taker credentials after restoring, and reopen access only after these checks succeed.

Open the **backup** app.
Locate the backup you want to restore in the backup list.



Click the **download button** next to your backup.

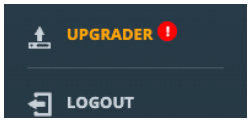
- Connect to your web server.
- Go to your OASYS root folder, and remove all files (including any hidden files and/or directories).
- Copy your downloaded backup archive ZIP file to your OASYS root.
- Extract the ZIP archive. In the OASYS 3.6 full-backup routine, the application entry point is archived as index.bkp. After extraction, check for this file and restore it to index.php before re-opening the application. If the package contains a different entry-point arrangement, verify it with the package provider.

- In your preferred database management software (or via your organization's DBA), remove the database which was being used for your OASYS instance.
- Re-create the database which was just removed, and leave empty (or request this operation to be done by your DBA).
- There will be a single .sql file in the OASYS root after extracting the archive — restore the contents of this file back into your database using your database software, or command line *(e.g., mariadb -u oasysuser -p my_oasys_db_name < backup.sql)*, or request this operation to be done by your DBA.
- Remove the .sql file from the root of your OASYS application directory after confirming that the system has been restored successfully **(this part is critical and should not be skipped, otherwise, future snapshot and backup operations can be adversely affected)**.

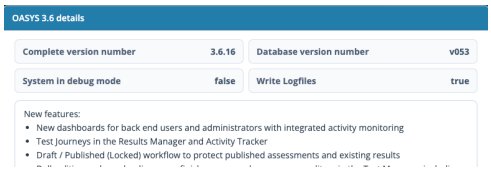
Chapter 8

System Upgrader

8.1 System Information



The currently installed OASYS version is displayed on the bottom of the main menu, right above the logout button.
Click the version number for detailed version information.



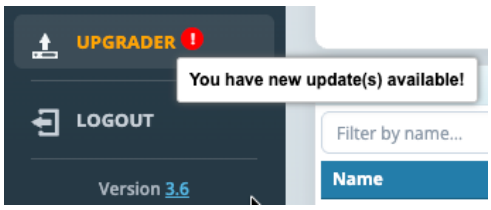
The popup screen shows the **complete version number**. The complete version identifies the installed release and patch level.
The OASYS database is versioned. As it doesn't change with every upgrade, the version number is different from the software version number.
System in debug mode and **write logfiles** are system settings. If the values aren't as intended by you, you can change them in the settings manager.
New features shows a list of changes which were introduced with the currently installed OASYS version.



Return to the popup screen after the upgrade.
The complete version number should have changed.
The OASYS database version only changes if the upgrade includes a database change.
System in debug mode and **write logfiles** should be as intended by you. The upgrade shouldn't change these settings.
Check the **new features** to be informed about changes in the software.
The **AGPL v3 License** button opens the AGPL license in a new browser window.

8.2 Update Notifications

The upgrader app is the tool to keep an OASYS installation on the newest available version.



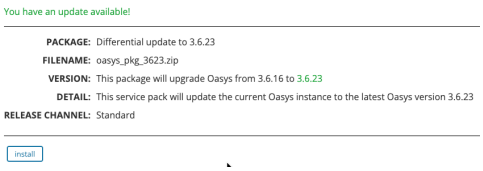
OASYS checks the update server for new versions. If there is an update available, you'll see a red exclamation mark close to the menu entry.

Custom Modules

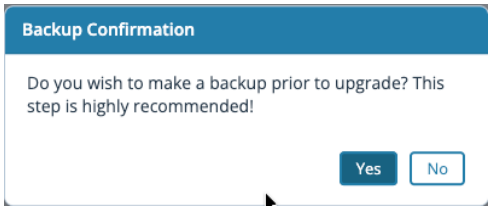
If you have custom modules installed, your OASYS installation is set up with a custom installer.
This does not affect the use of the upgrader!
The upgrader will automatically recognize your custom modules and integrate newer versions of your custom modules in the update.
If there are currently no versions of your custom modules which are compatible with the new OASYS version, the update will be rejected.

8.3 Performing Upgrades

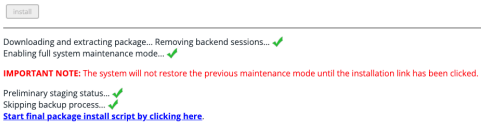
Review release notes and custom-module compatibility, schedule downtime, and download a full backup with matching encryption configuration before upgrading. Container deployments use their deployment process rather than the in-app upgrader. Click the **UPGRADER** button in the main menu.



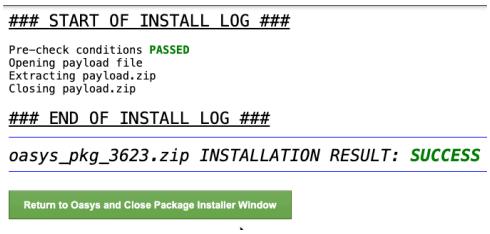
If there is an update, you'll see detailed information. Click **install** to start a precheck.



You will be prompted to perform a backup. When done, the backup will be available in the **backup** app.



If the precheck was passed without any errors, click the link below the check to perform the upgrade.



You will be directed to the installation log screen. Click the button below to return to OASYS.

8.4 Post-Upgrade Verification

Confirm the software and database versions, run system and file system checks, and inspect warnings. Test authentication, a representative assessment including media and result saving, result export, and backup creation. Check debug and logging settings and reopen access only after verification.

Chapter 9

Appendix

9.1 Required Privileges for the OASYS Database User

- SELECT
- INSERT
- UPDATE
- DELETE
- CREATE
- DROP
- ALTER
- INDEX
- LOCK TABLES
- EXECUTE
- CREATE VIEW
- SHOW VIEW